

Sparse Congruence-Preserving Perturbations of Dyadic Series

What gap statistics can tell us about Erdős Problem #251

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ABSTRACT

We show that a convergent dyadic series with nonnegative integer coefficients can attain an interval of values under sparse nonnegative perturbations. One set of permitted indices works for every target, with arbitrarily slowly growing correction bounds and eventual congruences for coefficients and partial sums. For prime gaps, the construction also preserves empirical blocks of growing length and the asymptotic size of the cumulative positions. Those positions need not be prime, and the irrationality question remains open.

1 Introduction

Let $p_0 = 2, p_1 = 3, \dots$ be the primes and put $g_n = p_{n+1} - p_n$. Erdős asked whether

$$\Pi = \sum_{n \geq 0} \frac{p_n}{2^{n+1}}$$

is irrational [1, p. 94][2, p. 62] [3, p. 103]. Summation by parts gives $\Pi = 2 + \sum_{n \geq 0} g_n 2^{-(n+1)}$, with convergence justified in Section 4. We study which properties of the gaps survive when the latter sum is changed to a prescribed value.

Our main result applies to any convergent dyadic series with nonnegative integer coefficients. We choose a sparse set of indices on which to add corrections. These indices are fixed before the target value, as are the cutoffs beyond which any prescribed modulus divides both the individual corrections and their partial sums. The possible values contain an interval. For prime gaps we can, in addition, retain the distributions of blocks of length $o(\log \log X)$ sampled in $[X, 2X)$, while the cumulative positions remain asymptotic to $n \log n$. Thus the irrationality question for the actual primes is a distinct target: the construction supplies comparison integer sequences without ensuring that their cumulative positions are prime.

We use a standard interval-covering argument for series with finite choices. Fridy's generalised-base lemma [5, p. 194] treats bounded digits and decreasing weights. Cr-marić and Kovač [17, Lemma 4] give the finite-choice form used here, and Kovač and

**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

Tao [6, Lemma 5.1] use analogous intervals of reciprocal choices. In our construction the choices must also respect sparsity, positivity and cumulative congruences. We arrange this by varying two adjacent corrections whose ordinary total is fixed. For example, the pairs $(0, 6), (2, 4), (4, 2), (6, 0)$ at indices $n, n + 1$ contribute 6, 8, 10, 12 divided by 2^{n+2} , although each pair has total 6. A correction at $n - 1$ first fixes the cumulative residue. The proof chooses the spacing of these triples so that the later choices still cover the gaps between the current ones.

Throughout, $\mathbb{N} = \{0, 1, \dots\}$, intervals of indices contain integers, and $\log$ is natural unless a base is displayed. A set $S \subseteq \mathbb{N}$ has *upper Banach density zero* if

$$\lim_{H \rightarrow \infty} \sup_{u \in \mathbb{N}} \frac{|S \cap [u, u + H)|}{H} = 0.$$

For integers $X, m \geq 1$, let $\mu_{a,X,m}$ denote the distribution of $(a_n, \dots, a_{n+m-1})$ when n is uniform on $[X, 2X)$. Blocks are unnormalised and counted with multiplicity. We use $d_{\text{TV}}(\mu, \nu) = \sup_B |\mu(B) - \nu(B)|$.

Proposition 1.1 (sparse changes preserving congruences). *Let $a : \mathbb{N} \rightarrow \mathbb{N}$ satisfy $A = \sum_{n \geq 0} a_n 2^{-(n+1)} < \infty$, let $K \in \mathbb{N}$, and let $f : \mathbb{N} \rightarrow \mathbb{R}$ tend to $+\infty$. There exist a set $S \subseteq [K, \infty)$ of upper Banach density zero and a nondegenerate interval $I \subset (A, \infty)$ such that, for every $r \in I$, there is an integer correction $e : \mathbb{N} \rightarrow \mathbb{N}$ satisfying*

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$$\text{supp } e \subseteq S, \quad e_n \leq f(n) \text{ eventually}, \quad \sum_{n \geq 0} (a_n + e_n) 2^{-(n+1)} = r.$$

For each integer $q \geq 1$ there is a cutoff N_q , independent of r , such that $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for all $n \geq N_q$. For every $\varepsilon > 0$, the choice $f(n) = (\log(n + 3))^\varepsilon$ can be made with

$$|S \cap [X, 2X)| = O_\varepsilon\left(\frac{X}{\log \log X}\right), \quad d_{\text{TV}}(\mu_{a,X,m}, \mu_{a+e,X,m}) \leq \frac{m|S \cap [X, 2X + m)|}{X}.$$

In particular the distance tends to zero for every integer-valued $m = m(X) \geq 1$ with $m(X) = o(\log \log X)$, uniformly over target values.

The support of e may depend on r and occupy only part of S . Polynomially growing nonnegative integer sequences satisfy the convergence hypothesis, whereas $a_n = 2^n$ does not. The allowance may grow as slowly as $\log \log(n + 3)$. A bounded allowance would force every correction to vanish: a modulus exceeding the bound makes e eventually zero, and its constant cumulative sum must then be divisible by every positive integer.

We prove the proposition in Section 2 and apply it to prime gaps in Section 3. The remaining sections give the actual-tail identity and the finite separation tests that a proof of irrationality would have to establish. The [companion's literature discussion](#) contains the further comparisons, including finite subsums and other dyadic coefficient sequences.

2 Proof of Proposition 1.1

Proof. Choose centres $n_j, j \geq 0$, after an initial index $n_{-1} \geq K$, and write $s_j = n_j - n_{j-1} \geq 4$. We shall choose positive integers M_j with $M_j \mid M_{j+1}$ and put $D_j = 2^{s_j} - 1$. At the j th

triple let C_j be the total correction before $n_j - 1$. Starting with $C_0 = 0$, set

$$c_j = (-C_j) \bmod M_j \quad (0 \leq c_j < M_j), \quad C_{j+1} = C_j + c_j + M_j D_j.$$

For a digit $0 \leq d_j \leq D_j$, define

$$e_{n_j-1} = c_j, \quad e_{n_j} = M_j d_j, \quad e_{n_j+1} = M_j(D_j - d_j), \quad (1)$$

and set $e_n = 0$ at the other indices. The triples are disjoint and each has total $c_j + M_j D_j$. Consequently every C_j and c_j is determined before any digits are chosen. Since $M_j \mid C_j + c_j$, the cumulative sum is divisible by M_j from index n_j through the end of this triple.

Suppose that $q \mid M_j$. At every later triple, $q \mid C_j$ and $q \mid M_j$ imply $q \mid c_j$, and both entries of the pair are divisible by q . It follows by induction, including the zero entries between triples, that $q \mid e_n$ and $q \mid \sum_{i < n} e_i$ for $n \geq n_j$. The initial residue correction c_j lies at $n_j - 1$, before this cutoff. Thus it remains to choose the moduli so that every fixed q eventually divides them, and to ensure that the weighted choices fill an interval.

For a possibly nonmonotone allowance, put

$$h(n) = \inf_{m \geq n} \min(f(m), m).$$

This is finite, nondecreasing and tends to infinity, with $h(n) \leq n$. Choose n_{-1} so that $h(n_{-1}) \geq 32$, and recursively set

$$k_j = \max\{k \geq 2 : k!2^{k+2} \leq h(n_{j-1})\}, \quad M_j = k_j!, \quad s_j = k_j + 2, \quad n_j = n_{j-1} + s_j.$$

The maximum exists because $2!2^4 = 32$ and factorials tend to infinity. Moreover, k_j is nondecreasing and tends to infinity, so the required divisibility holds and $s_j \rightarrow \infty$. Every entry in (1) is at most $M_j 2^{s_j} \leq h(n_{j-1})$. At a coordinate n of this triple it is therefore at most $f(n)$ and at most $n_{j-1} \leq n$. In particular all the weighted corrections are summable. Let S be the union of the triples. For any fixed R , all but finitely many centres are at least R apart. An interval of length H contains at most $3(H/R + 2)$ of their indices, apart from a fixed finite set. Taking the supremum over translates, then letting $H \rightarrow \infty$ and $R \rightarrow \infty$, proves that S has upper Banach density zero.

Write

$$w_j = M_j 2^{-n_j-2}, \quad \beta = \sum_{j \geq 0} (c_j 2^{-n_j} + D_j w_j), \quad F_j = \sum_{i \geq j} D_i w_i.$$

The weighted correction is $\beta + \sum_j d_j w_j$. Here $\beta > 0$, $0 < F_0 < \infty$ and $F_j \rightarrow 0$, by the bounds just proved. The choice of D_i gives

$$D_i w_i = M_i (2^{-n_{i-1}-2} - 2^{-n_i-2}).$$

For $i > j$ we have $M_i \geq M_j$, so summing this identity and taking a limit yields the overlap inequality

$$F_{j+1} \geq M_j 2^{-n_j-2} = w_j. \quad (2)$$

Hence the intervals

$$[dw_j, dw_j + F_{j+1}], \quad d = 0, \dots, D_j,$$

cover $[0, F_j]$. Given $x \in [0, F_0]$, we choose the digits successively so that $x - \sum_{i=0}^j d_i w_i \in [0, F_{j+1}]$. This remainder tends to zero, proving $x = \sum_j d_j w_j$. Taking $I = (A + \beta, A + \beta + F_0)$ proves the interval assertion. This is the finite-choice covering argument of [17, Lemma 4]. Notice that S, I and each cutoff n_j were fixed before x .

For the quantitative assertion, use the same triples with the schedule

$$s_j = \left\lfloor \frac{\varepsilon}{2} \log_2 \log(n_{j-1} + 3) \right\rfloor, \quad k_j = \max\{k \geq 2 : k! \leq (\log(n_{j-1} + 3))^{\varepsilon/4}\},$$

and again put $M_j = k_j!$, $n_j = n_{j-1} + s_j$ and $D_j = 2^{s_j} - 1$. Choose the initial index large enough that $s_j \geq 4$ and all these choices exist. Then

$$M_j 2^{s_j} \leq (\log(n_{j-1} + 3))^{3\varepsilon/4} \leq (\log(n + 3))^\varepsilon$$

at each coordinate n of the triple. This bound is also at most n after increasing the initial index. The summability, congruence and interval arguments therefore apply unchanged. Since $s_j \asymp_\varepsilon \log \log n_j$, the number of permitted indices in $[X, 2X)$ is $O_\varepsilon(X / \log \log X)$.

We couple the two block distributions by choosing the same starting index. A changed coordinate belongs to at most m of the sampled blocks, so the probability that the blocks differ is at most $m|S \cap [X, 2X + m)|/X$. This proves the stated total variation bound. When $m \leq X$, the interval $[X, 2X + m)$ is contained in $[X, 4X)$, and the support estimate on two dyadic intervals bounds the probability by $O_\varepsilon(m / \log \log X)$. The required uniform convergence follows. $\square$

The coupling also applies to tests depending on the starting index. If their absolute value is bounded by B_0 , their mean changes by at most $2B_0 m|S \cap [X, 2X + m)|/X$. For a function of the block alone, the bound is $2B_0 d_{TV}$. These are absolute errors: they give no relative estimate for an event whose probability tends to zero, nor a comparison of the complete infinite tails. The companion's [Section 2](#) also distinguishes upper Banach density from ordinary density.

3 Application to prime gaps

The elementary bound $p_n \leq 1250(n + 1)^4$ ensures convergence of the prime and gap series. For completeness, the central-binomial argument of Erdős [19, pp. 194–196] gives, for $m \geq 4$,

$$4^m < m \binom{2m}{m} \leq m(2m)^{\pi(2m)}.$$

The left inequality follows by induction. For the right inequality, the full power of any prime in $\binom{2m}{m}$ is at most $2m$. If $m = (n + 5)^4$ and $\pi(2m) \leq n$, these inequalities would give $4^m < m(2m)^n \leq 4^m$: with $x = n + 5$, use $x \leq 2^x$ and $n + 4(n + 1)x \leq 2x^4$ for the last inequality. Thus $p_n \leq 2(n + 5)^4 \leq 1250(n + 1)^4$. The full prime-power calculation is in [Appendix A of the companion](#).

Corollary 3.1 (a rational sum with the stated prime-gap statistics). *Let $p_0 = 2, p_1 = 3, \dots$ be the primes and $g_n = p_{n+1} - p_n$. Given $K \in \mathbb{N}$ and $0 < \varepsilon \leq 1$, there is $b : \mathbb{N} \rightarrow \mathbb{N}$ with rational dyadic sum such that $b_n = g_n$ for $n < K$, $b_n \geq g_n$, and $b_n - g_n \leq (\log(n+3))^\varepsilon$ eventually. For every fixed positive modulus, both the coefficients and the cumulative positions eventually retain their corresponding residues. The empirical distributions of unnormalised blocks have total variation distance tending to zero for lengths $o(\log \log X)$, and for every fixed nonzero $F \in \mathbb{Z}[x_0, \dots, x_k]$,*

$$|\{n < N : F(b_n, \dots, b_{n+k}) = 0\}| = o(N).$$

Moreover, with $P_n = 2 + \sum_{i < n} b_i$,

$$0 \leq P_n - p_n = O_\varepsilon\left(\frac{n(\log(n+3))^\varepsilon}{\log \log n}\right), \quad P_n \sim n \log n.$$

These positions are not asserted to be prime.

Lean proves this for every $\varepsilon > 0$, assuming Schläge-Puchta's Lemma 4 [18] only for the nonconcentration bound and the prime number theorem only for $P_n \sim n \log n$.

Proof. Apply Proposition 1.1 to $a = g$, choose a rational target $r \in I$, and put $b = g + e$. This gives the correction bounds, the eventual congruences and the comparison of growing blocks. Schläge-Puchta [18, Lemma 4] proved that the zero set of each fixed nonzero polynomial in a fixed block of prime gaps has density zero. A block of length $k+1$ changes only at a start in $\bigcup_{i=0}^k (S-i)$, which also has density zero. The conclusion for $F(b_n, \dots, b_{n+k})$ follows, regardless of the sizes of its values.

To estimate $P_n - p_n = \sum_{i < n} e_i$, first observe that $|S \cap [0, n)| = O_\varepsilon(n / \log \log n)$. Indeed, the indices below $\sqrt{n}$ contribute at most $\sqrt{n}$ points, and on $[\sqrt{n}, n)$ we sum the dyadic support bounds, whose lengths add to $O(n)$ and whose $\log \log X$ are comparable to $\log \log n$. Multiplying by the pointwise correction bound gives the stated error. For $0 < \varepsilon \leq 1$ this is $o(n \log n)$, so the prime number theorem [12, Chapter 6] gives $P_n \sim n \log n$. Finally, summing nonnegative terms in the opposite order, we obtain

$$\sum_{n \geq 0} \frac{P_n}{2^{n+1}} = 2 + \sum_{i \geq 0} b_i \sum_{n > i} 2^{-n-1} = 2 + \sum_{i \geq 0} \frac{b_i}{2^{i+1}} = 2 + r \in \mathbb{Q}.$$

□

For $\varepsilon > 1$ we may use exponent 1, which gives smaller corrections and the error $O(n \log(n+3) / \log \log n)$. Thus the conclusions extend to every $\varepsilon > 0$. For each fixed $y \geq 2$, eventual congruence modulo $y!$ also gives $\gcd(P_n, y!) = 1$ once $p_n > y$. The cutoff depends on y , so this excludes no growing range of possible prime divisors. Even a prime-valued sequence of positions could omit primes.

Land's draft [8, Theorem 2] proves conditional irrationality under a uniform Hardy–Littlewood hypothesis of the kind formulated by Kuperberg [7, Conjecture 1.3]. Ringer's draft [16] obtains conditional normality. Their quantitative prime-pattern hypotheses are stronger than the absolute block comparison above and are not assumed here. The companion's Section 3 compares the hypotheses and sampling conventions.

4 The prime series and its actual tails

Put $G = \sum_{n \geq 0} g_n 2^{-(n+1)}$. We now work with the actual primes and gaps throughout. The preceding polynomial bound justifies every series and limiting operation in the following identity.

Theorem 4.1 (prime-to-gap identity). *The actual prime and gap series satisfy $\Pi = 2 + G$. Their complete tails*

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$$T_N = \sum_{j \geq 1} g_{N+j} 2^{-j}$$

satisfy $T_0 = 2G - 1$ and $T_{N+1} = 2T_N - g_{N+1}$. Thus Π , G and T_0 have the same rationality status.

Proof. Finite summation by parts gives

$$\sum_{i=0}^n \frac{p_i}{2^{i+1}} = 2 + \sum_{i=0}^{n-1} \frac{g_i}{2^{i+1}} - \frac{p_n}{2^{n+1}}.$$

The endpoint tends to zero. Splitting off the first term of T_N gives its recurrence, and $g_0 = 1$ gives $T_0 = 2G - 1$. $\square$

Tao noted this reduction in the problem's forum discussion [15, 7 October 2025].

Corollary 4.2 (exact irrationality reformulation). *The prime-value dyadic series is irrational if and only if the prime-gap dyadic series is. Both series converge by the polynomial bound proved above; neither side is proved irrational.*

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Proof. Adding the rational number 2 preserves rationality and irrationality. $\square$

An arbitrary solution of the tail recurrence can differ from the actual tails by $C2^N$. The following boundary condition removes this ambiguity.

Lemma 4.3 (the boundary condition identifying a true tail). *Let $\sum_{j \geq 1} |a_j| 2^{-j} < \infty$ and $U_{N+1} = 2U_N - a_{N+1}$. Then $U_N = \sum_{j \geq 1} a_{N+j} 2^{-j}$ for every N if and only if $2^{-N} U_N \rightarrow 0$.*

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Proof. Iteration gives $2^{-N} U_N = U_0 - \sum_{j=1}^N a_j 2^{-j}$. The limit is zero exactly when U_0 is the full series. Subtracting its first N terms and multiplying by 2^N then gives the tail formula. Conversely the formula implies $2^{-N} U_N = \sum_{k > N} a_k 2^{-k} \rightarrow 0$ by absolute convergence. $\square$

4.1 Integral shifts

Let $U_{N+1} = 2U_N - a_{N+1}$ with $a_n \in \mathbb{Z}$, and write $D_h(N) = U_{N+h} - U_N$. Subtracting a_{N+1} leaves fractional parts unchanged, so modulo integers the recurrence is repeated doubling. This determines integral shifts from the denominator of U_0 alone.

Theorem 4.4 (exact rationality classification). *For a real integer-coefficient recurrence, the following are equivalent: $U_0 \in \mathbb{Q}$; $D_h(N) \in \mathbb{Z}$ for some $h \geq 1, N \geq 0$; and, for some fixed $h \geq 1$, $D_h(N) \in \mathbb{Z}$ at every sufficiently large N . More precisely, if $U_0 = u/(2^s d)$ is in lowest terms, with d odd, then*

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$$\text{den}(U_N) = 2^{\max(s-N,0)} d, \quad D_h(N) \in \mathbb{Z} \iff N \geq s \text{ and } d \mid 2^h - 1 \quad (h \geq 1).$$

Consequently U_0 is irrational exactly when every positive shift is nonintegral at every index, equivalently when for each fixed $h \geq 1$ there are arbitrarily late nonintegral shifts.

Proof. Iteration gives

$$U_N - 2^N U_0 \in \mathbb{Z}, \quad D_h(N) - 2^N(2^h - 1)U_0 \in \mathbb{Z}.$$

Since $2^N(2^h - 1) \neq 0$, one integral positive-length shift forces U_0 to be rational. If $U_0 = u/(2^s d)$ in lowest terms, multiplication by 2^N leaves denominator $2^{\max(s-N,0)} d$. Since $2^h - 1$ is odd, the second relation gives the asserted divisibility criterion. Euler's congruence supplies $d \mid 2^{q(d)} - 1$ when $d > 1$, and for $d = 1$ we take $h = 1$. Negating the resulting pointwise and eventual rationality criteria proves the last assertion. $\square$

The theorem requires neither a true-tail boundary condition nor prime-gap coefficients. The eventual periodicity is that of fractional parts, and need not hold for the full tails. For instance, the positive even coefficients $2, 4, 2, 4, \dots$ starting at $n = 1$ give tails $8/3, 10/3$ alternately. Every shift of length 1 is nonintegral and every shift of length 2 is zero. This explains why a single shift length is insufficient.

5 Small tail differences and finite tests

Fix $h \geq 1$ and put $D_N = T_{N+h} - T_N$ and $\delta_N = g_{N+h+1} - g_{N+1}$. Then $D_{N+1} = 2D_N - \delta_N$, where δ_N is even. Two differences of absolute value less than one can both be integral only if they vanish, in which case $\delta_N = 0$. More precisely, we have the following result.

Proposition 5.1 (two consecutive differences of absolute value less than one). *Let $D, D' \in \mathbb{R}$, $\delta \in 2\mathbb{Z}$ and $D' = 2D - \delta$. The conditions $|D| < 1$, $|D'| < 1$ and $\delta \neq 0$ hold exactly when, for some $s \in \{-1, 1\}$,*

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$$\delta = 2s, \quad \frac{1}{2} < sD < 1.$$

In that case $sD' \in (-1, 0)$ and both D and D' are nonintegral.

Proof. The bounds on D, D' imply $-3 < \delta < 3$, so a nonzero even δ equals $2s$ with $s \in \{-1, 1\}$. Substituting in $D' = 2D - 2s$ gives $1/2 < sD < 1$. Conversely this interval gives $sD' = 2sD - 2 \in (-1, 0)$. $\square$

For $\delta = 2$, the endpoints $D = 1/2, 1$ give $D' = -1, 0$ respectively. The zero triple $D = D' = \delta = 0$ shows why the mismatch is required.

Problem 5.2 (two small tail differences at arbitrarily large indices). *For every integer $h \geq 1$ and every cutoff N_0 , exhibit $N \geq N_0$ with*

$$|T_{N+h} - T_N| < 1, \quad |T_{N+h+1} - T_{N+1}| < 1, \quad g_{N+h+1} \neq g_{N+1}. \quad (3)$$

The problem is a sufficient condition for irrationality, by Theorem 4.4. Its three conditions must hold at the same index arbitrarily late for each h . Schlage-Puchta's lemma [18, Lemma 4], applied to $x_h - x_0 \pm 2$, shows that the eligible indices with $\delta_N = \pm 2$ already have density zero. Separate infinite sets of small-difference and mismatch witnesses would therefore not suffice.

5.1 Finite separation from the integers

Choose a majorant $M(n) \geq g_n$ with $\sum_n M(n)2^{-n} < \infty$, and put

$$S_{h,N,L} = \sum_{j=1}^L (g_{N+h+j} - g_{N+j})2^{-j}, \quad R_{h,N,L}(M) = \sum_{j>L} (M(N+h+j) + M(N+j))2^{-j}.$$

Then $|D_N - S_{h,N,L}| \leq R_{h,N,L}(M)$. A general real majorant need not give an effective remainder bound, but the polynomial choice below does.

Proposition 5.3 (finite separation criterion). *If for every $h \geq 1$ and every cutoff N_0 there are $N \geq N_0, L \geq 1$ with*

$$\text{dist}(S_{h,N,L}, \mathbb{Z}) > R_{h,N,L}(M), \quad (4)$$

then Π is irrational.

Proof. The distance of D_N from $\mathbb{Z}$ is at least $\text{dist}(S_{h,N,L}, \mathbb{Z}) - R_{h,N,L}(M) > 0$. Apply Theorem 4.4. $\square$

Remark (what arbitrary truncation does not gain). Whenever $|D - S_L| \leq R_L \rightarrow 0$, nonintegrality of D is equivalent to $\text{dist}(S_L, \mathbb{Z}) > R_L$ for some L . Indeed, for $\delta = \text{dist}(D, \mathbb{Z}) > 0$, take $R_L < \delta/2$ and use the 1-Lipschitz property of distance. Unrestricted depth therefore gives an exact reformulation, not an additional analytic estimate. At a prescribed logarithmic depth one must also control the margin from the integers as N varies; a shrinking remainder alone does not do so. The companion, Appendix D.2, gives an explicit example distinguishing refinement at a fixed value from enclosures of changing values.

For an explicit test, take $M(n) = 1250(n+2)^4$ and $P(x) = x^4 + 8x^3 + 36x^2 + 104x + 150$. Summing the identity $2P(x) = (x+1)^4 + P(x+1)$ with dyadic weights gives $\sum_{j \geq 1} (x+j)^4 2^{-j} = P(x)$ and hence

$$R_{h,N,L}(M) = 1250 2^{-L} (P(N+h+L+2) + P(N+L+2)).$$

If $\delta_N = 2s$ and $|QD_N - A| \leq B$, with $A \in \mathbb{Z}, Q > 0, B \geq 0$ integers, the inequalities

$$2sA - Q > 2B, \quad Q - sA > B$$

put sD_N in $(1/2, 1)$. The numerator A is signed. At $h = 1, N = 2, L = 40$, one has $\delta_2 = g_4 - g_3 = -2$ and

$$Q = 2^{40}, \quad A = -662838684750, \quad B = 11764181250, \quad s = -1.$$

Here $A/Q = S_{1,2,40}$ and $B/Q = R_{1,2,40}(M)$, and the two strict margins are 202637379224 and 424908761776. The [companion's integer certificates](#) give the prime-index calculation and remainder derivation. This finite pair leaves the quantifiers in Problem 5.2 unresolved.

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6 Further consequences and questions

The interval in Proposition 1.1 can lie in $(A, A + \eta)$ for any $\eta > 0$. Start after $K' \geq K$ with $(K' + 1)2^{-K'} < \eta$ and use $e_n \leq n$ from the proof. Then

$$0 < \sum_n e_n 2^{-(n+1)} \leq (K' + 1)2^{-K'} < \eta.$$

This bounds the dyadic change, without bounding the ordinary total.

6.1 Nonperiodic coefficients

Rational binary expansions have eventually periodic digits in $\{0, 1\}$. Integer coefficient sequences have more freedom, as the identity

$$\sum_{i=0}^{n-1} \frac{i-1}{2^{i+1}} = -\frac{n}{2^n} \rightarrow 0$$

shows. Its coefficients are unbounded, with first coefficient -1 . For a positive even example, put $c_n = 2(n^2 + 4n + 2)$ and $U_n = 2(n+4)^2$. We have $U_{n+1} = 2U_n - c_{n+1}$ and $2^{-n}U_n \rightarrow 0$, so Lemma 4.3 yields

$$\sum_{j \geq 1} c_j 2^{-j} = 32, \quad \sum_{n \geq 0} c_n 2^{-(n+1)} = 18.$$

The coefficients increase strictly and $c_{n+1} - c_n = 4n + 10$ is never ± 2 . The [companion's Section 8](#) gives further examples, including a bounded nonperiodic sequence in Appendix D. None of these examples constructs consecutive primes.

Proposition 6.1 (prime gaps do not become periodic). *For every positive h , the actual consecutive-prime-gap sequence is not eventually periodic with period h .*

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Proof. An eventually periodic sequence is bounded. For each $m \geq 2$, the $m - 1$ consecutive integers $m! + 2, \dots, m! + m$ are composite, so the consecutive primes on either side have a gap of at least m . $\square$

The results on bounded prime gaps [13], clusters of each fixed size [11, 20], and extreme large gaps [4] leave the signed weighted-tail windows above uncontrolled. This remains so with Polymath's bound $\liminf_n g_n \leq 246$ [20, Theorem 1.4(i)]. At a logarithmic truncation depth the explicit remainder tends to zero, but the finite sum must still be separated from the relevant boundary by a larger margin. For the actual prime series, nonintegral shifts are needed arbitrarily late for every positive length. Problem 5.2 would supply them. The perturbation theorem concerns the insufficiency of the specified congruences and block statistics, and leaves possible arguments using primality or quantitative rare-event estimates untouched.

7 Sources and further comparisons

Formal proofs. A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text. The linked Lean declarations use Lean 4 [9] and mathlib [10]. The formal proof of Proposition 1.1 changes one coordinate at each stage, whereas the proof here uses triples. Both fix the permitted set, the interval and the congruence cutoffs before choosing the target. For Corollary 3.1, the formal statement covers every $\varepsilon > 0$ and assumes Schläge-Puchta’s lemma only for nonconcentration and the prime number theorem only for the asymptotic $P_n \sim n \log n$. The source links record formal support for the linked statements, rather than verification of the different printed proofs. The [companion’s Appendix G](#) retains the complete source index at this note’s original pin, including the finite telescoping identities and their nonperiodicity consequences. Its [Section 3](#) retains the broader subsum and automatic-sequence comparisons.

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